Hearing the Substrate: Primordial Gravitational Waves from the Pre-Spatial Phase Transition

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1. The Ring

The Big Bang in BST is a phase transition: the pre-spatial state (fully saturated channel, all 137 slots occupied everywhere, no emergent geometry) nucleated into the spatial state (available channel capacity, circuit propagation, emergent 3D geometry). This transition released energy as the system fell from the high-energy saturated configuration to the low-energy spatial configuration.

The transition rang the substrate like a struck bell. The energy propagated across the Koons substrate as gravitational waves — ripples in contact density spreading outward from the nucleation point. These primordial gravitational waves carry information about the geometry of the transition: the shape of the nucleation event, the symmetry of the initial break, and the critical exponents of the phase transition on D^5_{IV}.

The electromagnetic echo of this ring is the cosmic microwave background — thermal radiation from the hot plasma that formed as the spatial phase cooled. The gravitational wave echo is different. Gravitational waves couple to the contact density directly, not through electromagnetic circuits. They propagate through the substrate itself. They are the substrate's own vibration.

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2. The Echo

The substrate is S^2 — a closed surface. If the phase transition nucleated at a single point, the transition wavefront propagated outward across the sphere in all directions. But S^2 is finite and has no boundary. The wavefront eventually reaches the antipodal point and converges — focused by the topology of the sphere into a concentrated echo.

The echo is a second ring, arriving at the antipode after one half-crossing time of the substrate. It is not a repetition of the original signal. It is the original signal focused and inverted by the spherical topology — amplified at the convergence point, carrying the signature of having traversed the full half-sphere.

The wavefront does not stop at the antipode. It passes through the convergence point, re-diverges, propagates back across S^2, and converges again at the original nucleation point. Then to the antipode again. Then back. The ring echoes back and forth across the closed substrate, each traversal fainter than the last as cosmic expansion dilutes the energy and the growing substrate increases the path length.

The result is a series of echoes:

Echo	Path	Relative amplitude	Information carried
Primary ring	Nucleation → observer	1 (reference)	Phase transition energy, critical exponents
First echo	Nucleation → antipode → observer	Reduced by expansion	Substrate diameter, topology
Second echo	Full circuit of S^2	Further reduced	Substrate curvature, damping rate
n-th echo	n half-crossings	Exponentially damped	Expansion history over n crossings

The spacing between echoes gives the diameter of S^2 — the size of the Koons substrate at the time of the transition. The damping rate gives the expansion history — how much the substrate grew between echoes. The spectral shape of each echo gives the critical exponents of the phase transition on D^5_{IV}.

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3. More Than One Ring

If the phase transition nucleated at multiple points on S^2 simultaneously, each nucleation produces its own ring. The gravitational wave background becomes a superposition of rings from different nucleation events.

The number of nucleation points depends on the thermodynamics of the transition:

Single nucleation (strongly first-order transition). One bubble of spatial phase forms and expands to fill the substrate. One primary ring. The universe began at one point on S^2. The gravitational wave background is a single spectral template with antipodal echoes.

Double nucleation. Two bubbles form at different points on S^2 and expand toward each other. Two primary rings with different timings and directions. The bubbles collide when their expanding wavefronts meet, producing a collision boundary — a ring-shaped topological defect on S^2 where the two spatial-phase regions merged. This defect may be observable as a specific pattern in the CMB or as a localized gravitational wave feature.

Multiple nucleation (weakly first-order transition). Many bubbles form simultaneously across S^2, expand, and merge. Many primary rings. The gravitational wave background is a complex superposition with a characteristic statistical structure. The boundaries where bubbles collided form a network of topological defects — domain walls, cosmic strings — distributed across the substrate. These defects are an additional source of dark matter in BST (see Working Paper Section 27.1).

Continuous nucleation (second-order transition). The transition occurs everywhere simultaneously with no distinct nucleation events. No discrete rings. The gravitational wave background is smooth with a spectrum determined entirely by the critical exponents. No topological defects from bubble collisions.

The distinction between these scenarios is encoded in the gravitational wave spectrum:

Scenario	Spectral signature	Defect content
Single nucleation	Single spectral template + echoes	None
Double nucleation	Two templates, different phases	One collision ring
Multiple nucleation	Statistical superposition	Network of defects
Continuous transition	Smooth, featureless spectrum	None

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4. Resonant Modes of the Substrate

The closed S^2 substrate has resonant modes — specific angular frequencies at which gravitational waves constructively interfere with their own echoes. These are the normal modes of the sphere, characterized by the spherical harmonic degree l:

$$f_l = \frac{c}{R_{S^2}} \sqrt{l(l+1)}$$

where R_{S^2} is the radius of the substrate at the time of the transition. The lowest mode ($l = 1$) is the dipole — one side of the substrate compressed while the other is rarefied. The quadrupole ($l = 2$) is the first mode that produces gravitational waves in the standard sense.

If the phase transition excited specific modes preferentially — which depends on the nucleation geometry — the primordial gravitational wave spectrum would show peaks at the corresponding frequencies, redshifted by the subsequent expansion. These peaks encode the topology of S^2 the way the resonant frequencies of a drum encode its shape.

BST predicts spectral features; inflation predicts a featureless spectrum. Standard slow-roll inflation produces a nearly scale-invariant primordial gravitational wave spectrum — no preferred frequencies, no peaks, no features. BST's phase transition on a closed substrate produces a spectrum with features determined by the substrate resonant modes and the nucleation geometry.

This is a clean observational discriminant between BST and inflation.

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5. The Physical Temperature Scale and the NANOGrav Prediction

5.1 T_c from BST Bubble Geometry

The phase transition temperature is set by the scale of a single channel bubble. In BST, the fundamental bubble has radius $R_s = 137\,\lambda_e$ (Working Paper Section 10.2), where $\lambda_e = \hbar/(m_e c)$ is the electron Compton wavelength. The critical temperature at which the substrate transitions from the fully-saturated pre-spatial phase to the spatial phase is:

$$T_c = \frac{130.5\,\hbar c}{k_B R_s}$$

The prefactor 130.5 comes from the critical heat capacity at the transition (Working Paper Section 15.1). Substituting $R_s = 137\,\lambda_e$:

Quantity	Value
$R_s = 137\,\lambda_e$	3.743×10^{-12} m
T_c	0.487 MeV $= 5.65 \times 10^9$ K
Cosmic time at T_c	3.1 seconds after the Big Bang
Epoch	Start of Big Bang Nucleosynthesis

The BST phase transition — the moment spatial geometry nucleated — occurred 3.1 seconds after the Big Bang, at the start of the BBN epoch.

This is not a coincidence to be explained: the electron sets both the mass scale ($m_\mu/m_e = (24/\pi^2)^6$) and the thermodynamic scale ($R_s = 137\,\lambda_e$) of the theory. The BBN epoch is precisely where the substrate first supported stable nucleon circuits.

5.2 The BST Timeline (0 to 3.1 Seconds)

In standard ΛCDM, the first 3.1 seconds contain the electroweak transition, quark-gluon plasma, QCD confinement, and $e^\pm$ annihilation — all described as particle processes. **In BST, there are no particles in the first 3.1 seconds.** The entire pre-spatial era is substrate dynamics:

Time	Temperature	BST description
$t = 0$	—	Substrate (D^5_{IV}) always existed; no singularity
$t \ll 3.1$ s	$\gg T_c$	Pre-spatial phase: all $\sim 10^{123}$ channels occupied, $C_v \approx 330{,}000$, no stable circuits = no particles
Pseudo-transition ~100 GeV	$\sim 10^{15}$ K	Effective EW symmetry breaking: pre-spatial sub-transition in substrate dynamics
Pseudo-transition ~150 MeV	$\sim 10^{12}$ K	Effective QCD confinement: pre-spatial color-channel sub-transition
$T \approx 2$ MeV	$\sim 2 \times 10^{10}$ K	Neutrino decoupling epoch; zero-mode channel propagation quanta decouple
$t = 3.1$ s	$T_c = 0.487$ MeV	**BST phase transition = the Big Bang**: first stable circuits, first particles, spatial geometry nucleates, GW ring emitted
$t > 3.1$ s	$< T_c$	Spatial phase: standard BBN, recombination, structure formation

The pre-spatial era dissolves two classical fine-tuning problems without inflation:

- **Horizon problem:** The substrate is fully connected in the pre-spatial phase (no spatial distances, no causal horizons). Homogeneity is set before spatial geometry forms.
- **Flatness problem:** D^5_{IV} is compact. The substrate topology is flat by construction — $k = 0$ is not a tuned initial condition but a geometric fact.

5.3 The NANOGrav Prediction

A first-order phase transition at temperature T_c produces a gravitational wave background peaking at the frequency (redshifted to today):

$$f_{\text{peak}} \simeq 1.9 \times 10^{-5}\,\text{Hz} \times \frac{T_c}{1\,\text{GeV}} \times \left(\frac{g_*}{100}\right)^{1/6}$$

With $T_c = 4.87 \times 10^{-4}$ GeV and $g_* = 10.75$ (photons $+e^{\pm}+$ 3 neutrinos at the BBN epoch):

$$\boxed{f_{\text{peak}} \approx 6.4\,\text{nHz}}$$

The NANOGrav 15-year dataset (2023) detected a stochastic gravitational wave background at 1–100 nHz, consistent with a power-law spectrum. The BST prediction of $f_{\text{peak}} \approx 6$–$9$ nHz falls directly in the detected band:

Source	Frequency
BST prediction ($T_c = 0.487$ MeV, sound waves)	≈ 6.4 nHz
BST prediction ($T_c = 0.487$ MeV, turbulence)	≈ 9.1 nHz
NANOGrav 2023 detected band	1–100 nHz

The spread from 6.4 to 9.1 nHz reflects the two dominant GW production mechanisms (sound waves vs. MHD turbulence). The NANOGrav spectral peak is observed at $\sim$ 25–30 nHz; BST sits at the low end of the band. Precise matching requires computing the transition duration β/H_* and efficiency factor κ from the BST partition function dynamics.

5.4 Transition Strength

The phase transition strength parameter is $\alpha = \Delta V/\rho_{\text{rad}}$, where ΔV is the latent heat and ρ_{rad} is the radiation energy density at T_c.

From the BST partition function computation (Notes: BST_PartitionFunction_Analysis.md), the transition has heat capacity $C_v \approx 330{,}000$ at T_c — three orders of magnitude above a weakly first-order electroweak transition. This implies $\alpha \gg 1$: the BST transition is **ultra-strong**.

The gravitational wave energy density for $\alpha \gg 1$:

$$\Omega_{\text{GW}} h^2 \sim 10^{-5} \times \left(\frac{\alpha}{1+\alpha}\right)^2 \times (H_* R_*)^2 \approx 10^{-7}$$

(estimating $H_* R_* \sim 0.01$ for a strong transition at the BBN scale). This is within the sensitivity of current pulsar timing arrays and is potentially the dominant contribution to the NANOGrav 2023 background.

6. Observational Prospects

6.1 CMB B-Mode Polarization

Primordial gravitational waves imprint a specific polarization pattern (B-modes) on the cosmic microwave background. The amplitude is characterized by the tensor-to-scalar ratio r. The spectral shape is characterized by the tensor spectral index n_t and any features (peaks, oscillations) in the tensor power spectrum.

LiteBIRD (launch planned ~2032) will measure r to a sensitivity of ~ 0.001 and characterize the B-mode spectrum at large angular scales.

CMB-S4 (operations planned ~2030s) will measure the B-mode spectrum at smaller angular scales with high sensitivity.

BST prediction: The B-mode spectrum has features at angular scales determined by the resonant modes of S^2 at the time of the phase transition. The feature spacing encodes the substrate size. The feature amplitude encodes the nucleation geometry.

6.2 Pulsar Timing Arrays

Pulsar timing arrays (NANOGrav, EPTA, PPTA, IPTA) are sensitive to gravitational waves at nanohertz frequencies — much lower than LIGO's band. In 2023, these arrays reported evidence for a stochastic gravitational wave background at nanohertz frequencies.

BST interpretation: The nanohertz background may include a component from the primordial ring, redshifted into the nanohertz band by cosmic expansion. If the background has spectral features (peaks at specific frequencies rather than a smooth power law), this would support the substrate resonance picture.

6.3 LISA

LISA (launch planned ~2035) will detect gravitational waves in the millihertz band. Primordial gravitational waves from the phase transition, if they exist in this band, would appear as a stochastic background with the spectral features predicted by BST.

6.4 Direct Detection of Echoes

The most dramatic prediction: if the substrate is small enough that the echo crossing time is less than the age of the universe, we might detect discrete echoes — repeated bursts of gravitational wave energy arriving at regular intervals. The interval gives the substrate circumference. The damping gives the expansion rate.

This requires the substrate to be finite and not too much larger than the observable universe. If the substrate is vastly larger than the observable universe, the echo crossing time exceeds the age of the universe and no echoes have arrived yet. The substrate size is currently unknown and is one of the quantities the primordial gravitational wave spectrum could measure.

7. Connection to Existing BST Predictions

The primordial gravitational wave predictions connect to several existing BST results:

CMB anomalies (Working Paper Section 15.3). The large-angle CMB anomalies (hemispherical power asymmetry, Cold Spot, low-multipole alignment) may be imprints of the substrate topology. The gravitational wave resonant modes and the CMB anomalies should share the same angular scale structure — both are determined by the geometry of S^2. A consistent angular scale between B-mode features and temperature anomalies would be strong evidence for the substrate topology.

Topological defects as dark matter (Working Paper Section 27.1). If the phase transition nucleated at multiple points, the collision boundaries between expanding spatial-phase bubbles are topological defects. These defects contribute to the dark matter budget. The primordial gravitational wave spectrum encodes the number and distribution of nucleation points, which determines the defect density, which determines the topological defect contribution to dark matter.

Phase transition critical exponents (Working Paper Section 15.1). The spectral shape of the primordial ring is determined by the critical exponents of the pre-spatial to spatial phase transition on D^5_{IV}. These same critical exponents determine the CMB spectral index n_s and the tensor-to-scalar ratio r. All three observables — n_s, r, and the gravitational wave spectral features — are derived from the same phase transition and must be mutually consistent.

The growing manifold (Working Paper Section 24). The primordial ring propagates across the committed substrate. Each echo traverses the substrate as it existed at the time of that echo's crossing. The series of echoes samples the expansion history of the committed manifold at discrete intervals — each echo is a snapshot of the substrate size at the time it completed its crossing.

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8. New Thesis Topics

1. **Primordial gravitational wave spectrum from D^5_{IV} phase transition:** Compute the tensor power spectrum from the critical exponents of the pre-spatial to spatial transition, including resonant mode structure from S^2 topology
2. **Substrate echo detection:** Calculate the echo spacing and damping rate as functions of substrate size and expansion history; assess detectability with LiteBIRD, CMB-S4, LISA, and pulsar timing arrays
3. **Multiple nucleation signatures:** Compute the gravitational wave spectrum from N simultaneous nucleation events on S^2 and the resulting topological defect network; compare to observed stochastic gravitational wave backgrounds
4. **CMB anomaly — B-mode correlation:** Predict the angular scale relationship between temperature anomalies and B-mode polarization features if both arise from S^2 substrate topology
5. **NANOGrav spectral fit:** Fit the BST prediction ($f_{\text{peak}} = 6$–9 nHz, $\alpha \gg 1$) to the NANOGrav 15-year power-law background; compute the transition parameters β/H_* and κ from the BST partition function
6. **Entropy injection and BBN light elements:** The BST phase transition at $T_c = 0.487$ MeV injects latent heat ($C_v = 330{,}000$) into the plasma after proton-neutron freeze-out ($T \approx 0.8$ MeV). Compute the entropy dilution factor and its effect on the ^{4}He, D, and ^{7}Li yields; the

lithium-7 problem (observed ^{7}Li is $\sim 3\times$ below standard BBN prediction) may reflect BST entropy injection during BBN

7. **Pre-spatial era effective description:** Map the BST sub-transitions (EW at ~ 100 GeV, QCD at ~ 150 MeV, neutrino decoupling at ~ 2 MeV) onto pre-spatial substrate dynamics; derive effective temperatures and transition parameters from BST partition function analysis

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9. Summary

The pre-spatial phase transition rang the Koons substrate. The ring propagates as primordial gravitational waves. The closed topology of S^2 produces echoes — the ring bouncing back and forth across the finite substrate, each traversal carrying information about the substrate size and the expansion history.

If the transition nucleated at one point, we hear one ring with echoes. If it nucleated at multiple points, we hear multiple rings — the substrate boiled rather than cracked, and the collision boundaries between expanding spatial-phase bubbles are topological defects that contribute to dark matter.

The BST phase transition temperature is $T_c = 0.487$ MeV, placing the birth of spatial geometry at $t = 3.1$ seconds — the start of the BBN epoch. This is determined from first principles by $R_s = 137\,\lambda_e$ and the BST critical heat capacity.

The gravitational wave ring from this transition peaks at $f_{\text{peak}} \approx 6$–$9$ nHz. **NANOGrav 2023 detected a stochastic gravitational wave background at 1–100 nHz — the BST prediction is in the observed band.** With transition strength $\alpha \gg 1$ ($C_v = 330{,}000$), the BST transition is ultra-strong and should produce a large GW signal.

The first 3.1 seconds in BST are the pre-spatial era. There are no particles, no singularity, no inflation. The horizon and flatness problems are resolved by substrate topology. At $t = 3.1$ s, the first stable circuits form — the first protons, neutrons, and electrons — and the standard cosmological history begins.

BST also predicts that the entropy injected at T_c may explain the cosmological lithium-7 problem: the $\sim 3\times$ discrepancy between standard BBN predictions and observed ^{7}Li abundances. The latent heat ($C_v = 330{,}000$) dilutes the effective baryon-to-photon ratio during the ^{7}Be production phase, reducing ^{7}Li by the required factor.

The primordial gravitational wave spectrum encodes the birth geometry of reality. Inflation predicts a featureless spectrum. BST predicts features at specific frequencies set by the substrate geometry and the resonant modes of S^2. LiteBIRD, CMB-S4, LISA, and pulsar timing arrays will measure which prediction is correct.

We may hear the substrate directly. If we hear more than one ring, we know the universe didn't begin with a bang. It began with a boil. And the echoes are still arriving.

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